stmforest package description

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v20250918

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These are the forest definitions for `stmlatex`. It is build upon the forest package.

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1. Usage - in the preamble

1.1. Load the whole `stmforest` package

1.1.1. Description

This is an interface package which loads `forest` and definitions commonly required throughout document creation.

By default the package loads

- `stmforestbase.sty`
- `stmforestlibraries.sty`
- `stmforeststyles.sty`

See subsubsection 1.1.2 for options to change the default package behavior.

1.1.2. Options

Option *libraries* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the standard libraries commonly required.

```
\usepackage[libraries=true|false]{stmforest}
```

`libraries=true` is the default. It is used in case `libraries=false` is not set explicitly.

Option *styles* This is a boolean option. Expected values are either `true` or `false`. It controls whether to load the predefined `forest` styles.

```
\usepackage[styles=true|false]{stmforest}
```

`styles=true` is the default. It is used in case `styles=false` is not set explicitly.

1.2. `stmforestlibraries`

This package contains standard libraries commonly required in the creation of `forest` trees.

1.3. `stmforeststyles`

This package contains styles commonly required in the creation of `forest` trees.

2. Examples

2.1. Directory tree

A simple example of how to create a nice directory tree from the styles defined in `stmforeststyles`. The result is shown in Figure 1.

3. Known issues

...

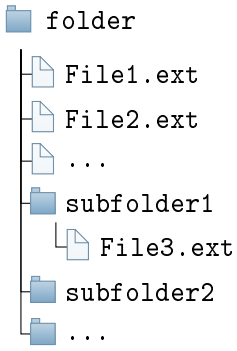


Figure 1: Directory tree example

A. The code

A.1. stmforest.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                               %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to all stm forest definitions
% Based upon the forest package:
%   https://ctan.org/pkg/forest
%
% Usage
%  - Preamble:
%    - \usepackage{stmforest}
%
% Revisions: 2024-11-08 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                  __/|__
%                                 /_/_/_/
%            www.dlr.de/sy/en       |/ DLR
%
% Copyright (C) 2024-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                              %
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmforest}[2024/11/08 STMs custom LaTeX forest
    definitions]

% ---------------------------
% Options
% ---------------------------

% For options
\@ifpackageloaded{kvoptions}{}{\RequirePackage{kvoptions}}%

% Option group
\SetupKeyvalOptions{%
  family=stmforest,%
  prefix=stmforest@,%
  setkeys=\kvsetkeys,%
}

% Libraries
\DeclareBoolOption[true]{libraries}

% Styles
\DeclareBoolOption[true]{styles}

% Process options
\ProcessKeyvalOptions{stmforest}

% ---------------------------
% Modules
% ---------------------------

% Load the base package
\@ifpackageloaded{stmforestbase}{}{%
  \RequirePackage{stmforestbase}%
}%

% Libraries
\ifstmforest@libraries
```

```
  \RequirePackage{stmforestlibraries}
\fi

% Styles
\ifstmforest@styles
  \RequirePackage{stmforeststyles}
\fi

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```

A.2. stmforestbase.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This is a interface to stm base forest package
% Based upon the forest package:
%   https://ctan.org/pkg/forest
%
% Usage
%  - Preamble:
%    - \usepackage{stmforestbase}
%
% Revisions: 2024-11-08 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
%                                  __/|__
%                                 /_/_/_/
%            www.dlr.de/sy/en       |/ DLR
%
% Copyright (C) 2024-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a comment about what it's used for
\ProvidesPackage{stmforestbase}[2024/11/08 STMs custom LaTeX forest definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{forest}{}{\RequirePackage{forest}}%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading this file. LaTeX will ignore anything after this line.
\endinput
```

A.3. stmforestlibraries.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
% This file includes unit definitions.
% Based upon the forest package:
%   https://ctan.org/pkg/forest
%
% Usage
%  - Preamble:
%    - \usepackage{stmforestlibraries}
%
% Revisions: 2024-11-08 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
%
```

```
%                                    __/|__
%                                   /_/_/_/
%            www.dlr.de/sy/en         |/ DLR
%
% Copyright (C) 2024-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a comment about what it's used for
\ProvidesPackage{stmforestlibraries}[2024/11/08 STMs custom LaTeX forest library definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{forest}{}{\RequirePackage{forest}}%

% ---------------------------
% Libraries
% ---------------------------

\useforestlibrary{edges}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading this file. LaTeX will ignore anything after this line.
\endinput
```

A.4. stmforeststyles.sty

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Header                             %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
%
```

```
% This is a interface to all stm forest definitions
% Based upon the forest package:
%   https://ctan.org/pkg/forest
%
% Usage
%  - Preamble:
%    - \usepackage{stmforeststyles}
%
% Revisions: 2024-11-08 Martin Raedel <martin.raedel@dlr.de>
%                       Initial draft
%
% Contact:   Martin Raedel,  martin.raedel@dlr.de
%            DLR Lightweight Systems
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%                                 __/|__
%                                /_/_/_/
%            www.dlr.de/sy/en      |/ DLR
%
% Copyright (C) 2024-... DLR Lightweight Systems
%
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% Content                            %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Declare that this style file requires at least LaTeX version 2e.
\NeedsTeXFormat{LaTeX2e}

% Provide the name of your page, the date it was last updated, and a
    comment about what it's used for
\ProvidesPackage{stmforeststyles}[2024/11/08 STMs custom LaTeX forest
    style definitions]

% ---------------------------
% Package
% ---------------------------

\@ifpackageloaded{forest}{}{\RequirePackage{forest}}%
\@ifpackageloaded{stmcolors}{}{\RequirePackage{stmcolors}}%

% ---------------------------
% Styles
% ---------------------------

\definecolor{folderbg}{RGB}{124,166,198}
```

```
\definecolor{folderborder}{RGB}{110,144,169}


\newlength\ZZZTempSize
\setlength\ZZZTempSize{4pt}
\tikzset{%
  folder/.pic={%
    \filldraw [draw=folderborder, top color=folderbg!50, bottom color=
    folderbg] (-1.05*\ZZZTempSize,0.2\ZZZTempSize+5pt) rectangle ++(.75*\
    ZZZTempSize,-0.2\ZZZTempSize-5pt);
    \filldraw [draw=folderborder, top color=folderbg!50, bottom color=
    folderbg] (-1.15*\ZZZTempSize,-\ZZZTempSize) rectangle (1.15*\
    ZZZTempSize,\ZZZTempSize);
  },
  file/.pic={%
    \filldraw [draw=folderborder, top color=folderbg!5, bottom color=
    folderbg!10] (-\ZZZTempSize,.4*\ZZZTempSize+5pt) coordinate (a) |- (\
    ZZZTempSize,-1.2*\ZZZTempSize) coordinate (b) -- ++(0,1.6*\ZZZTempSize
    ) coordinate (c) -- ++(-5pt,5pt) coordinate (d) -- cycle (d) |- (c) ;
  },
}
\forestset{%
  declare autowrapped toks={pic me}{},
  declare boolean register={pic root},
  pic root=0,
  pic dir tree/.style={%
    for tree={%
      folder,
      font=\ttfamily,
      grow'=0,
    },
    before typesetting nodes={%
      for tree={%
        edge label+/.option={pic me},
      },
      if pic root={
        tikz+={
          \pic at ([xshift=\ZZZTempSize].west) {folder};
        },
        align={l}
      }{},
    },
  },
  pic me set/.code n args=2{%
```

```
    \forestset{%
      #1/.style={%
        inner xsep=2\ZZZTempSize,
        pic me={pic {#2}},
      }
    }
  },
  pic me set={directory}{folder},
  pic me set={file}{file},
}

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
% That's it                          %
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

% Finally, we'll use \endinput to indicate that LaTeX can stop reading
    this file. LaTeX will ignore anything after this line.
\endinput
```